Smart Medicine Availability and Janaushadhi Equivalent Identification System with Real-Time Price Transparency

***Abstract*—Finding required generic medicine can be difficult when it is not available at nearby pharmacies or when customers are unaware of alternative medicines.This paper presents a Smart Medicine Availability And Janaushadhi Equivalent Identification System With Real-Time Price Transparency System that brings medicine search, prescription recognition and pharmacy stock information into one platform.User can search medicine by entering its name or by uploading an image of prescription.When a prescription is uploaded, the system uses a pre-trained machine learning model to extract medicine names.The extracted names are then matched with records in the medicine database.when the medicine is identified by the system it provides its composition, generic verison of the medicine, janaushadhi alternative, nearby janaushadhi recommendation, stock and price information.Pharmacy owners can use this system to maintain stock quantities, price, batch number, expiry dates.This data is used to identify pharmacies where required medicine or its generic alternative is in the stock. The system also helps user to find nearby pharmacies using location based pharmacy search. By connecting medicine identification, generic alternative suggestion, janaushadhi stock and price comparison the proposed system provides a single platform for finding and comparing medicine related information.**

***Keywords*— Janaushadhi, medicine availability, generic medicines, prescription recognition, machine learning, pharmacy inventory, price comparison.**

I. Introduction

Getting the required medicine is not always easy, especially when it is not available at nearby. Patients often face difficulty to find the particular medicines, alternatives, particularly when medicines are known mainly by their brand names. The lack of pharmacy stock information may require patients to visit multiple stores, while handwritten prescriptions can make accurate medicine identification more difficult to identify. At the pharmacy level, maintaining updated information about stock quantities, batches, expiry dates and prices is also necessary for inventory management.

The proposed Smart Medicine Availability and Janaushadhi Equivalent Identification System addresses these challenges by providing a platform connecting customers with pharmacy inventory services. The system uses an ML-based prescription recognition model to extract medicine names from prescription images and match them with records in a structured medicine database. Based on the identified medicine, the system provides composition details, generic and Janaushadhi alternatives, and price comparisons. It also uses pharmacy inventory information and location-based services to help users identify pharmacies where the required medicine or an available alternative may be found.

The proposed system integrates ML-based prescription recognition, medicine identification, generic and Janaushadhi recommendation, price comparison, pharmacy availability, and inventory management within a single platform. Pharmacy operators can manage medicine stock, batch details, expiry dates, prices and sales, while customers can access medicine information and locate the nearby pharmacies. By connecting medicine information with pharmacy-level inventory, the system aims to improve affordability, accessibility and transparency in medicine availability.

II. Related Work

Various studies have explored the use of machine learning to identify medicines from handwritten medical prescriptions. In [1], researchers investigated the extraction and classification of medicine names from prescription images. The work in [2] extended prescription recognition by converting the recognized information into audio, while [3] examined deep neural networks for interpreting doctors' handwritten prescriptions. CRNN-based recognition was explored in [4] to address the challenges associated with handwritten medical text. Related research has considered machine-learning-based prescription recognition [5], multilingual recognition [6], and the combination of neural networks with information retrieval [7]. The study in [8] further examined automated prescription interpretation in the context of patient safety. Together, these studies establish the relevance of prescription-image processing for medicine identification. However, identifying a medicine from a prescription is only the first step in helping patients obtain it. Users also need to know whether suitable alternatives exist, how their prices compare, and which nearby pharmacies have the required medicine available.

Beyond prescription recognition, researchers have investigated intelligent techniques for pharmacy management and medicine recommendation. The system in [9] combined computer vision with an automated medicine delivery robot, demonstrating the application of technology to pharmacy operations. In medicine recommendation, [10] explored interactions between medicine-related information through a heterogeneous graph, whereas DKINet [11] incorporated domain knowledge into the recommendation process. DAINet [12] investigated adaptive interactions for coordinated medication

recommendation, and [13] examined personalized recommendations using medical order information. Machine learning was also applied to drug recommendation in [14]. The study in [15] brought disease prediction, medication recommendation, and question-answering together in a deep learning-based medical system, while [16] addressed decision-making in hospital pharmacy environments. Although these approaches address medicine selection and pharmacy-related decisions, the proposed system has a different emphasis: helping users find an identified medicine, compare it with composition-based generic alternatives, and check its reported availability and price at nearby pharmacies.

Medicine availability also depends on how effectively pharmacies manage their inventory. Research in [17] examined AI-enhanced pharmacy inventory management, while [18] focused on inventory management and prediction. SupplyRx [19] addressed pharmaceutical inventory and supply monitoring, and [20] investigated machine-learning techniques for forecasting sales and inventory demand. Other work has considered the classification of hospital pharmaceutical inventory using ABC analysis and fuzzy classification [21], as well as cost optimization across pharmaceutical supply-chain levels [22]. More recent studies have explored CNN–LSTM-based demand forecasting [23] and machine-learning-based drug-dispensation forecasting in a public hospital [24]. Artificial intelligence applications in pharmacy management [25] and inventory management for small and medium-sized enterprises [26] further demonstrate the range of problems addressed by intelligent inventory systems. These studies focus on improving stock control, forecasting demand, reducing inventory-related problems, or optimizing supply operations. For a patient searching for a medicine, however, inventory information becomes more useful when it is connected directly to medicine identification, price comparison, and the location of pharmacies that report having the medicine in stock.

The reviewed literature covers prescription recognition [1–8], medicine recommendation and pharmacy management [9–16], and inventory and supply-chain management [17–26]. However, these areas are often addressed separately, leaving a gap in connecting medicine identification with affordable alternatives, price comparison, and nearby pharmacy availability. Existing approaches may not provide all these functions together, particularly for identifying Janaushadhi alternatives and checking pharmacy stock. This highlights the need for a more integrated approach that connects medicine information, affordability, and recorded pharmacy inventory to simplify the medicine search process.

Medicine affordability has also received attention through initiatives that promote access to lower-cost generic medicines. In India, the Pradhan Mantri Bhartiya Janaushadhi Pariyojana (PMBJP) aims to make quality generic medicines available at affordable prices through dedicated Janaushadhi Kendras [27]. However, identifying a suitable generic alternative and determining where it can be obtained may require users to consult different sources for medicine composition, price, and pharmacy availability. Price information alone does not establish whether a medicine is available at a nearby store, while identifying a generic alternative requires appropriate matching of medicine composition, strength, and formulation. Integrating medicine information, recorded prices, and pharmacy inventory can therefore help users explore affordable medicine options more conveniently. This motivates the inclusion of Janaushadhi alternative identification and location-based pharmacy search in the proposed system, alongside prescription-based medicine identification and inventory management.

III. PROPOSED METHODOLOGY

The proposed Smart Medicine Availability and Janaushadhi Equivalent Identification System integrates prescription-based medicine identification, generic alternative recommendation, price comparision and inventory management. Users can search by medicine name or by uploading a prescription image, from which a pre-trained machine learning model extracts medicine names. The identified medicines are matched with database to retrieve their composition and find suitable generic and Janaushadhi alternatives. The system compares prices, checks pharmacy stock records, and uses location information to identify nearby pharmacies . It consists of three modules: customer, pharmacy, and administration, connected through a common backend and database to manage medicine information, inventory, users, and pharmacy records.

A. System Architecture

The proposed system follows an architecture that integrates medicine identification, information retrieval, pharmacy availability checking and inventory management. It connects customers, pharmacy operators, and administrators, with each group has its own functions according to the role. The customer interface supports medicine searches and displays medicine and pharmacy details, while pharmacy operators maintain stock information used to determine medicine availability. Administrators manage system information and user access. All components communicate through a common backend and database, allowing information updated by one module to be accessed by other modules when required.

The architecture is organized into the following submodules.

a) Customer Module

The Customer Module provides the main interface through which users can search for medicines and access relevant information stored in the system. Customers can enter a medicine name manually or upload an image of a prescription. When a prescription image is uploaded, a pre-trained machine learning model identifies and extracts the medicine names from the image. The extracted names are then sent to the medicine database for further processing. Once a medicine is identified, the system retrieves its composition and searches the database for suitable generic and Janaushadhi alternatives. This helps customers explore different medicine options without having to search for each alternative separately.

The module displays the identified medicine and its available alternatives along with relevant price details by enabling

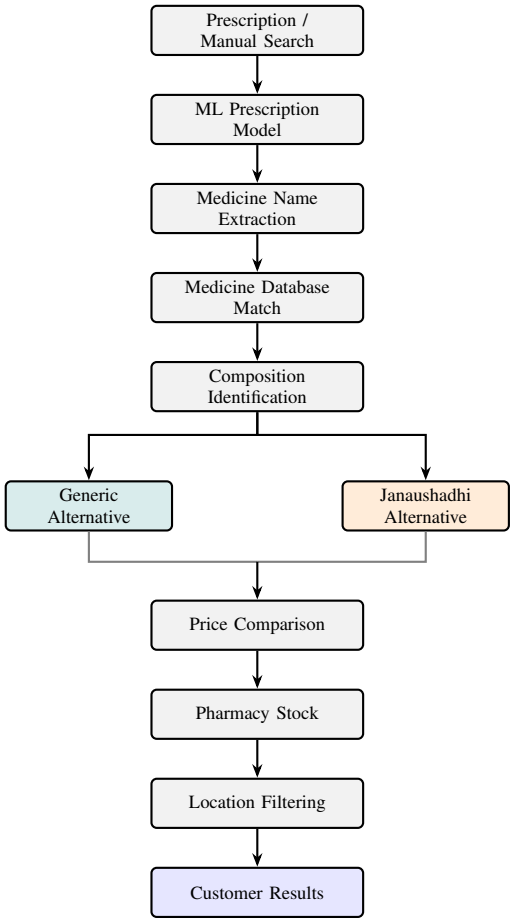


Fig. 1. Block diagram of the medicine alternative recommendation pipeline.

customers to compare the listed options. It also checks the recorded pharmacy inventory to determine whether the required medicine and its alternatives are available in the stock . Using location-based search, customers can identify nearby pharmacies that may have the required medicine in stock. By integrating manual medicine search, prescription-based identification, alternative discovery, price comparison, and pharmacy location services into a single interface, the Customer Module simplifies the medicine search process. However, the accuracy of the results depends on successful medicine identification, the completeness of the database, and the regular updating of pharmacy inventory information.

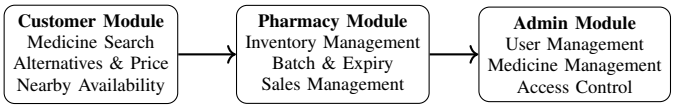


Fig. 2. Overview of the system modules.

b) Pharmacy Inventory Module

The Pharmacy Module enables pharmacy owners to manage the medicine inventory and maintain updated records in the system. After logging in through JWT-based authentication, pharmacy owners can add new medicines and update existing records, including medicine names, batch numbers, stock quantities, prices, and expiry dates. When a medicine is sold, its quantity can be updated to reflect the latest recorded stock. The module also provides low-stock alerts and expiry notifications, helping pharmacy owners identify medicines that require restocking or attention.

The module also supports sales history and report generation to help pharmacy owners to monitor their stock movement and pharmacy activities. All updated inventory and pricing details are stored in the database and used by the customer module during medicine searches. When customers search for a medicine or a suitable generic or Janaushadhi alternative, the system checks the recorded pharmacy inventory and displays relevant price and availability information. Location-based pharmacy search helps customers identify nearby pharmacies where the required medicine is listed as available. Thus, the Pharmacy Module connects inventory management with customer services, while the accuracy of displayed stock information depends on regular updates by pharmacy owners.

c) Administration and Authentication Module

The Admin Module manages the main activities of the platform, including user accounts, pharmacy owner approvals, medicine information, and system settings. The administrator can add, update, or remove medicine records, including medicine names, ingredients, and Janaushadhi alternatives. The module also supports the management of medicine categories and alternative mappings used by the customer and pharmacy modules, ensuring that the information maintained across the platform remains organized and consistent.

The administrator can also view reports related to user registrations, pharmacy activities, and inventory changes. In addition, system settings and access permissions can be managed according to user roles. The system uses JWT-based authentication and role-based access control to restrict access to different modules. After login, the user's role determines the functions they are permitted to access. Customers can search for medicines and view alternatives, while pharmacy owners can manage their inventory, prices, batch details, and expiry dates. Administrators have access to user management, medicine database management, and other administrative functions. This role-based approach helps protect sensitive operations and ensures that users can access only the features permitted for their roles.

B. System Implementation and Workflow

The proposed system integrates prescription recognition, medicine identification, generic and Janaushadhi alternative mapping, price comparison, pharmacy availability, and inventory management. Users can search for a medicine by name or upload a prescription image, from which a pre-trained machine learning model extracts medicine information. The identified medicines are matched with database records to find suitable generic and Janaushadhi alternatives and compare their prices. The system also checks pharmacy stock records and uses location-based services to help users find nearby pharmacies. Additionally, pharmacy operators can update stock quantities

and manage batch and expiry details, while administrators manage medicine and pharmacy information. These modules work together to help users identify affordable medicine options and access relevant availability information.

TABLE I
CONVENTIONAL MEDICINE SEARCH VS. PROPOSED SYSTEM

Feature	Conventional Approach	Proposed System
Medicine Search	Manual search	Centralized search
Prescription Processing	Manual identification	ML-based recognition
Generic Identification	Limited	Automated mapping
Janaushadhi Alternatives	Not integrated	Available
Price Comparison	Manual comparison	Automated comparison
Pharmacy Availability	Visit/call pharmacies	Inventory-based search
Nearby Pharmacy	Manual search	Location-based search
Stock Management	Separate records	Integrated inventory
Batch and Expiry	Manual tracking	Batch-level tracking

a) Medicine Identification

The medicine identification module accepts medicine names entered manually or prescription images uploaded by users. For prescription-based searches, a pre-trained machine learning model processes the uploaded image and extracts the names, which are then forwarded to the backend for further processing. The model is used for recognition and is not trained on the dataset. The extracted medicine names are normalized and matched with structured medicine records stored in MongoDB to retrieve available details such as the name, composition, strength, and price. For manual searches, the system directly performs database matching without requiring prescription processing. Both input methods follow the same identification process, providing a common source of information for subsequent operations. If a medicine cannot be matched reliably, its identity should be verified before displaying alternatives.

b) Medicine Alternatives and Price Comparison

After identifying a medicine, the system searches the available database for generic and Janaushadhi alternatives using the medicine's composition and other recorded attributes. The matching alternatives and their available prices are presented together to help users compare branded medicines, generic options, and Janaushadhi products. This approach makes price differences easier to understand without requiring users to search multiple sources separately. The comparison depends on the completeness of the stored medicine records, and the system displays only the information available in the database. Alternatives are provided for informational purposes and should not be treated as automatic substitutes for prescribed medicines, since factors such as strength and dosage form must also be considered. Users are advised to consult a qualified healthcare professional before changing their medication. By combining alternative identification with price comparison, the module helps users explore potentially more affordable medicine options.

Algorithm 1 Weighted Sum Based Inventory Prioritization

Require: Medicine inventory records
Ensure: Priority score and ranked inventory list
1: Define criteria: Stock Level, Demand, Expiry Risk, and Sales Rate
2: Assign a weight w_i to each criterion such that $\sum w_i = 1$
3: **for** each medicine m in inventory **do**
4: Normalize the value of each criterion
5: Calculate priority score:
6: $Score(m) = \sum_{i=1}^{n} w_i \times x_i$
7: Store the calculated score for medicine m
8: **end for**
9: Sort medicines in descending order of priority score
10: Generate inventory priority list
11: **return** ranked inventory list

c) Pharmacy and Inventory Management

The pharmacy and inventory management module connects medicine searches with the stock information maintained by participating pharmacies. Pharmacy operators can add medicines and update stock quantities, prices, batch numbers, and expiry dates through the inventory interface. These changes are sent to the FastAPI backend and stored in MongoDB, allowing the customer-side search to use the latest recorded inventory information. When a user searches for a medicine or alternative available, the system checks the stored pharmacy records and combines the results with the pharmacy location information to identify nearby options. A map-based interface displays pharmacy locations to help users find relevant outlets. Inventory availability depends on the accuracy and frequency of pharmacy updates and does not necessarily represent automatically detected physical stock. The inventory module also helps pharmacy operators maintain medicine records and monitor batch and expiry details. By integrating stock management, availability checking, and location-based pharmacy discovery, the system connects the medicine information with practical options to obtain the required products.

IV. RESULTS AND ANALYSIS

This section presents the results of the implementation of the proposed Smart Medicine Availability and Janaushadhi Recommendation System. The evaluation focuses on the identification and alternative recommendation, comparative price analysis, and pharmacy availability. The discussion examines how integrating these functions improves access to medicine-related information compared to approaches that require users to search for alternatives, compare prices, and locate pharmacies separately.

A. Medicine Identification and Alternative Recommendation

The proposed system accepts medicine names through manual entry or prescription images processed using a pre-

TABLE II
AVERAGE UNIT PRICE BY PROCUREMENT SOURCE

Procurement Source	Avg. Price (Rs.)
Generic Aadhar	12.68
Branded Medicines	11.17
Branded Generic Medicines	9.12
Generic Distributors	9.10
Dava India	4.88
Jan Aushadhi	2.40
Kerala Medical Services Corporation Limited	0.00

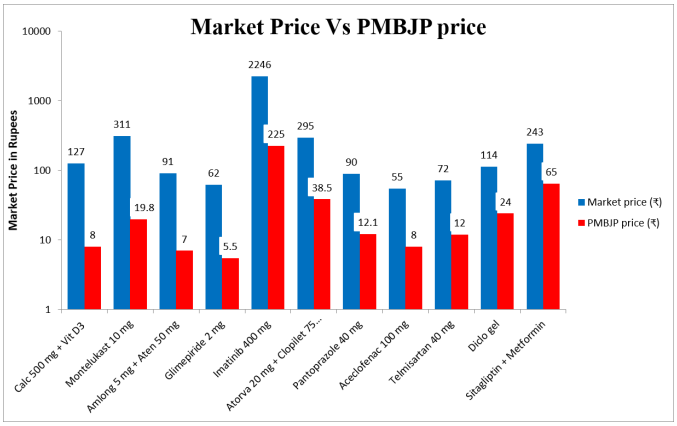


Fig. 3. Market Price vs PMBJP

trained machine learning model. The identified medicine names are matched with the integrated medicine dataset to retrieve relevant information, including composition, strength, dosage form, and available generic alternatives. The dataset is preprocessed to standardize medicine names, remove duplicate records, and normalize relevant fields, supporting consistent database matching. Following a successful match, the system retrieves the corresponding generic and Janaushadhi alternatives when suitable records are available. Compared to searching by medicine name alone, the use of composition and strength information provides additional criteria to identify relevant alternatives. However, the results depend on the recognition model, the quality of the prescription images, and the completeness of the medicine dataset. Incorrectly recognized names or missing records can prevent successful matching, and alternative recommendations require verification of formulation and strength before they can be considered clinically appropriate.

B. Comparative Price Analysis

The price comparison module enables users to examine recorded market prices alongside PMBJP prices for selected medicines. Fig. 3 presents the comparison between the two price categories, highlighting the cost differences in the selected records. The percentage price reduction is calculated using the market price as the reference value:

$$\text{Reduction } (\%) = \frac{\text{Market Price} - \text{PMBJP Price}}{\text{Market Price}} \times 100 \quad (1)$$

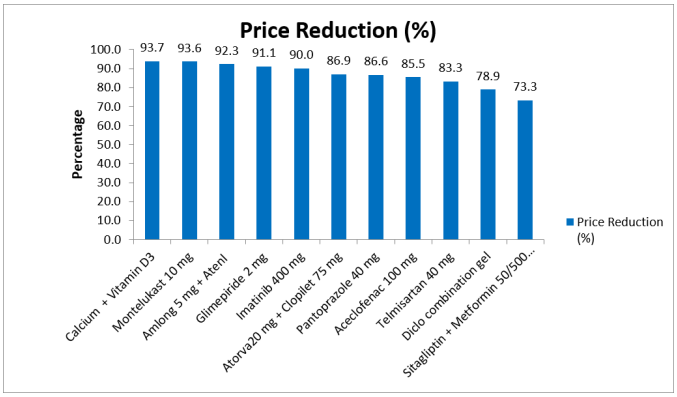


Fig. 4. Price Reduction

This analysis provides a clearer view of potential cost savings than displaying medicine prices individually. Unlike a price-only information source, the proposed system connects price information with medicine identification and alternative recommendation, allowing users to review available alternatives and their recorded prices within the same workflow. The comparison is based on the selected published price records and should not be interpreted as a guaranty of savings for all medicine. Differences in pack size, strength, formulation, and price-update date can affect the comparison. Therefore, the source and reference date of the prices should be considered when interpreting the results.

C. Pharmacy Availability and Integrated System Evaluation

The pharmacy module connects medicine records to pharmacy-specific inventory and location information. Pharmacy owners can update stock quantities and maintain relevant details, including batch numbers and expiry dates, when supported. The system uses the recorded quantities to determine the stock status and applies location-based filtering to help users identify nearby pharmacies or Janaushadhi Kendras. Compared with approaches that provide medicine information and price comparisons without linking them to pharmacy inventory, the proposed system also helps users identify locations where a medicine is listed as available. The integrated workflow connects medicine identification, alternative recommendation, price comparison, and pharmacy discovery, reducing the need to consult separate sources for each task. Nevertheless, the accuracy of availability information depends on the frequency and accuracy of inventory updates, and the displayed stock may differ from physical availability at the time of purchase. Further testing using verified prescription images, confirmed medicine matches, and physical stock checks to quantify recognition accuracy, matching reliability, and consistency of pharmacy availability information.

V. CONCLUSION

The Smart Medicine Availability and Janaushadhi Equivalent Identification System was developed to make it easier for people to find medicines, compare prices, and locate

nearby pharmacies. Instead of searching for medicine details, generic alternatives, prices, and pharmacy information separately, users can access these features through a single platform. The system accepts medicine names entered manually or extracted from prescription images using a pre-trained machine learning model. Then it matches the medicine with the available database records and suggests generic and Janaushadhi alternatives when suitable information is available. Integration of price comparison, pharmacy location, and inventory management also allows users to check where a medicine is listed as available, while pharmacy owners can maintain stock quantities, batch details, prices, and expiry dates. Thus, the main contribution of the system is to bring these related functions together in one place.

Although the system simplifies the search for medicines, its usefulness still depends on how accurately prescription names are identified, how complete the medicine database is, and how regularly pharmacies update their stock information. The availability displayed may not always reflect the actual stock in a pharmacy, and the suggested alternatives should be checked for suitability before use. In the future, the system can be improved by testing the prescription recognition model with a wider range of real prescriptions, improving medicine matching for unclear or incomplete names, and keeping medicine prices and pharmacy stock information more frequently updated. Further testing with users and pharmacy owners would also help identify areas for improvement and determine how effectively the system supports everyday medicine searches.

REFERENCES

[1] S. R. Edlabadkar and P. P. Joshi, "Accurate Medicine Recognition and Audio Conversion for Handwritten Prescriptions," in *Proc. IEEE ESCI*, 2024, pp. 1–6, doi: 10.1109/ESCI59607.2024.10497304.

[2] M. F. Uz Zaman, B. Rahman, A. Rubyat, N. Halder, A. Mahmud, A. Islam, and M. A. Amin, "Recognizing Medicine Names from Bangladeshi Handwritten Prescription Images using TrOCR," in *Proc. CEII*, 2024, pp. 55–59, doi: 10.1109/CEII65291.2024.00020.

[3] A. V. Sable, A. Patil, M. Rathi, and A. Shriwas, "Interpreting Doctor Notes Using Handwriting Recognition," 2024, doi: 10.22214/ijraset.2024.60663.

[4] N. M. Jyothi, J. B. Simha, and K. V. Kiran Kumar, "Word Prediction from Medical Prescription via Transfer Learning with Pre-trained CNN, GAN and BiLSTM Integration," in *Proc. NMITCON*, 2024, doi: 10.1109/NMITCON62075.2024.10698946.

[5] H. Wu, Y. Gan, M. Wei, D. He, et al., "Intelligent Pharmacy Management System Based on Computer Vision Technology and New Medicine Delivery Robot," in *Proc. ICCEA*, 2024, doi: 10.1109/ICCEA62105.2024.10604058.

[6] "OCR-Based Extraction of Expiry Dates and Batch Numbers in Medicine Packaging for Error-Free Data Entry," in *Proc. ICCPCT*, 2024, doi: 10.1109/ICCPCT61902.2024.10673325.

[7] G. K. Prashanth and H. S. Yashaswini, "Drug Recommendation System Using Machine Learning," in *Proc. ICSSS*, 2024, doi: 10.1109/ICSSS62373.2024.10561403.

[8] "Empowering Patient Care: A Novel Patient Drug Recommendation System Using Artificial Intelligence with Modified Learning Strategy," in *Proc. IEEE*, 2024, doi: 10.1109/ICCDS60734.2024.10560419.

[9] "Personalized Drug Information and Recommendation System Using Gradient Boosting Algorithm (GBM)," in *Proc. IEEE*, 2024, pp. 923–926, doi: 10.1109/SMART63812.2024.10882477.

[10] F. Gao, N. Zhao, Y. Chen, J. Guo, et al., "Personalized Prescription Recommendation Using Attention Over Medical Order Information," *IEEE Access*, vol. 12, 2024, doi: 10.1109/ACCESS.2024.3459080.

[11] F. Zhu, X. Zhang, B. Zhang, Y. Xu, and L. Cui, "Medicine Package Recommendation via Dual-Level Interaction Aware Heterogeneous Graph," *IEEE J. Biomed. Health Inform.*, vol. 28, 2024, doi: 10.1109/JBHI.2024.3361552.

[12] S. Liu, X. Wang, X. Zhao, and H. Chen, "DKINet: Medication Recommendation via Domain Knowledge Informed Deep Learning," in *Proc. IEEE BIBM*, 2024, doi: 10.1109/BIBM62325.2024.10821909.

[13] X. Zou, X. He, X. Zheng, W. Zhang, J. Chen, and C. Tang, "DAINet: Dual Adaptive Interaction Network for Coordinated Medication Recommendation," *IEEE J. Biomed. Health Inform.*, vol. 28, 2024, doi: 10.1109/JBHI.2024.3425833.

[14] S. Chairat, A. Sae-Ang, K. Suvirat, T. Ingviya, and S. Chaichulee, "Enhancing Medication Recommendation with Hierarchical Network and Patient Visit Histories," in *Proc. IEEE EMBC*, 2024, doi: 10.1109/EMBC53108.2024.10781496.

[15] A. Syed Musthafa, S. Bhuvaneshwari, S. S. Dharshini, U. Gobika, et al., "Enhancing Patient Safety with Machine Learning: Automating the Interpretation of Handwritten Medical Prescriptions," in *Proc. ICUIS*, 2024, doi: 10.1109/ICUIS64676.2024.10867175.

[16] K. U. Kiru, S. K. Pal, S. S. Ahmad, and I. M. Kabir, "Deep Learning-Enhanced Medical System for Disease Prediction, Medication Recommendation, and Intelligent Question-Answering," in *Proc. IEEE UPCON*, 2024, doi: 10.1109/UPCON62832.2024.10982945.

[17] "Medicine Authentication Based on Image Processing Using Convolutional Neural Networks," in *Proc. ICCAE*, 2024, doi: 10.1109/ICCAE59995.2024.10569752.

[18] "MediExchange: A Smart Web Portal to Buy and Sell Unused Medicines," in *Proc. IEEE ICSES*, 2024, doi: 10.1109/ICSES63445.2024.10763152.

[19] F. Kochakkashani, V. Kayvanfar, and R. Baldacci, "Innovative Applications of Unsupervised Learning in Uncertainty-Aware Pharmaceutical Supply Chain Planning," *IEEE Access*, vol. 12, 2024, doi: 10.1109/ACCESS.2024.3435439.

[20] "Sales and Inventory Demand Forecasting Using Machine Learning," in *Proc. ICETEMS*, 2024, pp. 173–178, doi: 10.1109/ICETEMS64039.2024.10964960.

[21] K. Poongothai, G. Devika, D. S. Brisila, and S. Yogesh, "Smart Retail Using Machine Learning for Demand Forecasting and Inventory Optimization," in *Proc. ICSES*, 2024, doi: 10.1109/ICSES63760.2024.10910874.

[22] V. Pavithra, H. M. Abbas, M. B. Manjula, P. K. Chidambaram, and S. Jothilakshmi, "Optimizing Inventory Management using Demand Forecasting based Ensemble Stacking Model with Hyperparameter Tuning," in *Proc. ICISCN*, 2025, doi: 10.1109/ICISCN64258.2025.10934488.

[23] D. Singh, H. Singh, S. Kumar, V. Garg, et al., "SupplyRx: Intelligent Pharmaceutical Inventory and Supply Monitoring System," in *Proc. CE2CT*, 2025, doi: 10.1109/CE2CT64011.2025.10939612.

[24] "Predictive Machine Learning Models for the Management of Pharmaceutical Drugs," in *Proc. IEEE*, 2025, Doc. 11014998.

[25] D. Abregu-Garcia, N. Camacho-Rondinel, and L. Romero-Varona, "Improving Order Fulfillment in Pharmaceutical Supply Chains Using Deep Reinforcement Learning," in *Proc. IEEE INTERCON*, 2025, doi: 10.1109/INTERCON67304.2025.11244663.

[26] G. K. Kumar, P. R. Aadith Krishna, N. M. Rithik, and V. Varun, "Smart Tracker: An Automated System for Tracking Expiry Dates of Groceries and Medicines," in *Proc. IEEE ICMSCI*, 2026, pp. 1685–1689, doi: 10.1109/ICMSCI67830.2026.11469409.

[27] Pharmaceuticals and Medical Devices Bureau of India (PMBI), "About Pradhan Mantri Bhartiya Janaushadhi Pariyojana (PMBJP)," [Online]. Available: https://janaushadhi.gov.in/about-pmbjb. [Accessed: Oct. 18, 2026].